

and non-cleavable linker strategies, reflecting different solutions to the same central problem of balancing stability with efficient payload delivery (Lin et al., 2025).

Approved non-cleavable linkers include Trastuzumab emtansine which uses the SMCC linker with DM1, while belantamab mafodotin uses a maleimidocaproyl linker with MMAF (Jang and Chang, 2025). In these systems, payload release depends on intracellular antibody degradation rather than a defined enzymatic linker cleavage event. This strategy can improve plasma stability and reduce premature release (Su et al., 2021; Lin et al., 2025).

Several approved ADCs use chemically cleavable linkers that respond to intracellular conditions. Gemtuzumab ozogamicin and inotuzumab ozogamicin use AcButDMH linkers with calicheamicin, historically described as acid-labile hydrazone-type systems designed to release payload in acidic endosomal/lysosomal compartments. Mirvetuximab soravtansine uses sulfo-SPDB, a disulfide-containing linker paired with DM4, making release sensitive to the intracellular reducing environment. Loncastuximab tesirine uses a PEG–Val–Ala–PABC linker with a pyrrolobenzodiazepine dimer payload, combining hydrophilic PEGylation with protease-cleavable peptide release (Jang and Chang, 2025).

Overall, the approved linker landscape differs mainly along four axes: cleavable versus non-cleavable release, trigger mechanism (protease, acid, reduction, hydrolysis) or antibody degradation payload compatibility, and bystander potential. Protease-cleavable peptide linkers dominate many modern solid-tumor ADCs because they can combine stable circulation with efficient lysosomal release. Non-cleavable linkers favor systemic stability. Hydrazone, disulfide and hydrolysable linkers provide alternative release mechanisms but can raise different stability and toxicity trade-offs.

Cell penetrating peptides

The cell membrane protects the interior of every living cell with remarkable precision. It permits small nutrients, ions, and selected signals to pass, yet it excludes many molecules that researchers would like to deliver for diagnosis, imaging, and therapy. For decades, this barrier limited the therapeutic use of large or fragile cargoes, including proteins, peptides, nucleic acids, siRNA, nanoparticles, and imaging agents. Cell-penetrating peptides changed that story by offering a way to carry biological cargo across the plasma membrane and into the cell.

Researchers first recognized this possibility in 1988 while studying the HIV-1 transactivator of transcription protein, Tat. They observed that Tat could enter mammalian cells and influence viral gene expression,

suggesting that part of the protein possessed an unexpected membrane-crossing ability (Guo et al., 2016; Gori et al., 2023). This finding challenged the assumption that proteins required classical receptors, channels, or disruptive delivery methods to gain intracellular access. It also raised an important question: could short protein regions themselves behave as transport modules?

In 1991, Alain Joliot and colleagues, working at the Centre National de la Recherche Scientifique in Paris, showed that a 60-amino-acid peptide from the *Drosophila melanogaster* Antennapedia homeodomain could enter mammalian nerve cells. This peptide retained the biological activity expected of an Antennapedia homeodomain because it bound a homeobox consensus DNA sequence from the Hox-1.3 promoter in a gel-shift assay.

This work then led researchers to ask which part of the 60-amino-acid homeodomain was responsible for cell entry. Later studies showed that uptake was lost when researchers mutated the C-terminal region of the homeodomain, directing attention to the third α -helix in that region. In 1994, Derossi, Joliot and colleagues chemically synthesized shorter peptides from this third helix and tested whether they could still cross cell membranes. They found that 20-amino-acid and 16-amino-acid fragments retained membrane-crossing ability, whereas shorter 15-amino-acid fragments did not. The active 16-amino-acid sequence, RQIKIWFQNRRMKWKK, later became known as penetratin or Antp.

Antennapedia homeodomain:

RKRGRQTYTRYQTLELEKEFHFNRYLTTRRRRIEIAHALCLTERQIKI
WFQNRRMKWKKEN

Antp/penetratin, derived from the third helix of the Antennapedia homeodomain:

RQIKIWFQNRRMKWKK

Cell-penetrating peptides, or CPPs, now describe a diverse family of short peptides, commonly around 5–40 amino acids long, that can move across cellular membranes and transport attached or associated cargo. Many CPPs carry a positive charge at physiological pH because they contain arginine and lysine residues. These amino acids help them interact with negatively charged membrane components, including phospholipids and

heparan sulphate proteoglycans; positively charged CPPs bind to negatively charged heparan sulphate proteoglycans on the cell surface, which helps start cellular uptake. Other CPPs use amphipathic or hydrophobic structures to associate with lipid bilayers and promote internalization. (Gori et al., 2023; Guo et al., 2016).

Scientists classify CPPs in several overlapping ways. Some arise from natural proteins, such as Tat, penetratin, VP22, lactoferrin-derived peptides, and toxin-derived sequences. Others combine natural and synthetic elements to form chimeric peptides, while fully synthetic CPPs allow researchers to design charge, stability, amphipathicity, and targeting properties from the start. Researchers also group CPPs by physicochemical behavior, including cationic, amphipathic, hydrophobic, and even anionic peptides (Gori et al., 2023).

CPPs enter cells through more than one route. Some cross membranes directly through energy-independent mechanisms such as pore formation, membrane thinning, carpet-like disruption, or inverted micelle formation. Others enter through energy-dependent endocytosis, including macropinocytosis, clathrin-mediated endocytosis, caveolae-mediated endocytosis, and related pathways. The exact route depends on the CPP sequence, cargo size, concentration, incubation time, temperature, cell type, and membrane composition. Because many CPP-cargo complexes enter endosomes first, endosomal escape remains one of the central challenges in CPP-mediated delivery (Gori et al., 2023; Yang, Liu and Sha, 2025).

Their promise lies in their ability to move difficult cargoes into cells. Researchers have used CPPs to deliver anticancer drugs, proteins, oligonucleotides, siRNA, plasmid DNA, imaging probes, quantum dots, nanoparticles, and vaccine antigens. In cancer research, CPPs have gained attention because they can improve intracellular delivery and tumor penetration. Recent work highlights their role in tumor immunotherapy, where CPPs may support cytokine therapy, adoptive cell therapy, cancer vaccines, immune checkpoint blockade, and tumor-targeted nanocarriers. Peptides such as iRGD show promise because they can improve tumor tissue penetration and help therapeutic agents reach poorly accessible tumor regions (Yang, Liu and Sha, 2025).

Despite this promise, CPPs still face important barriers before broad clinical translation. Many CPPs lack tissue specificity, and uncontrolled uptake may increase off-target effects. Some peptides show limited stability in biological fluids, while others struggle with poor endosomal release after cellular uptake. Researchers therefore continue to improve CPP design through cyclisation, stapling, retro-inverso modification, dendrimeric structures, tumor-targeting motifs and nanoparticle conjugation. These strategies aim to increase stability, reduce toxicity, enhance selectivity and

improve cargo release inside target cells (Gori et al., 2023; Yang, Liu and Sha, 2025).

DaxibotulinumtoxinA-lanm, marketed as DAXXIFY®. DAXXIFY is a botulinum toxin type A product that acts as an acetylcholine-release inhibitor and neuromuscular-blocking agent. It is approved for the temporary improvement of moderate-to-severe glabellar lines and for the treatment of cervical dystonia in adults. Unlike many botulinum toxin formulations that use human serum albumin as a stabilizing component, DAXXIFY is formulated with RTP004, a proprietary 35-amino-acid peptide excipient.

RTP004 contains highly basic, Tat-like protein-transduction domains. Its reported sequence, RKKRRQRRRGKKKKKKKKKKKKKKKKKKKKGRKKRRQRRR, includes repeated RKKRRQRRR motifs, which closely resemble the classic HIV-Tat cell-penetrating sequence. These positively charged domains flank a polylysine core, giving RTP004 strong cationic character like many CPPs.

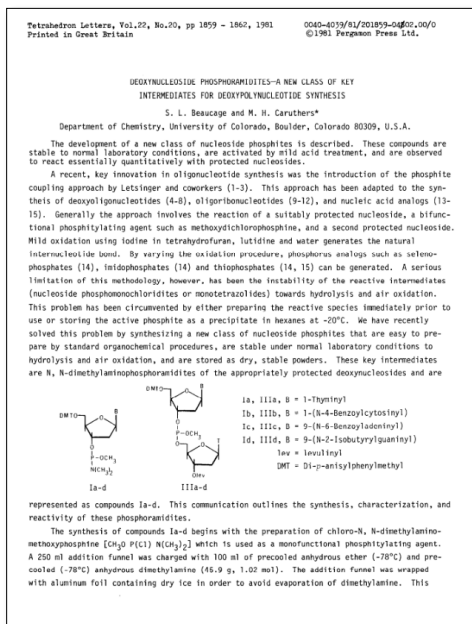
Ionis Pharmaceuticals and Bicycle Therapeutics entered a collaboration in 2021 to test whether Bicycle's constrained peptide technology could improve the targeted delivery of Ionis' oligonucleotide therapeutics. Drug delivery is especially difficult for skeletal and cardiac muscle, where systemically administered antisense oligonucleotides and siRNAs must leave the bloodstream, cross tissue barriers and enter cells before they can engage intracellular targets. Conventional oligonucleotides show limited potency in these tissues.

A 2025 study from Ionis and Bicycle identified BCY17901, a small bicyclic peptide of approximately 2 kDa that binds human transferrin receptor 1 (TfR1), a recycling receptor expressed on skeletal muscle cells and cardiomyocytes. BCY17901 was conjugated to ASO and siRNA to improve receptor-mediated uptake into muscle and heart. In human TfR1 knock-in mice, BCY17901 conjugation improved ASO potency by 5.2–8.4-fold in quadriceps, 6.9–7.6-fold in gastrocnemius and at least 5.8-fold in heart compared with unconjugated ASOs. For siRNA, BCY17901 conjugation improved potency by 5.5-fold in quadriceps, 4.7-fold in gastrocnemius and 2.7-fold in heart compared with a lipid-conjugated siRNA control.

The approach translated into nonhuman primates, despite BCY17901 having about 10-fold lower affinity for cynomolgus monkey TfR1 than for human TfR1. BCY17901-conjugated MALAT1 ASO reduced target RNA by 71% in quadriceps and 63% in heart. BCY17901-conjugated HPRT siRNA reduced target RNA by 86% in quadriceps and 75% in heart. The study estimated that delivering a hypothetical 3 mg/kg oligonucleotide-equivalent dose would require roughly 250–300 mg of a Bicycle–ASO

conjugate, compared with about 6000 mg for a monoclonal antibody–ASO conjugate.

ASO Synthesis

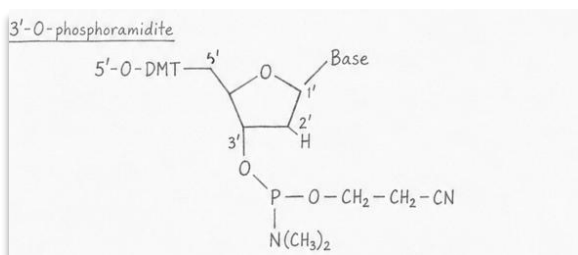


The development of antisense oligonucleotides depended on major advances in chemical synthesis that made production practical, reproducible, and ultimately automated. Modern ASO synthesis emerged from a series of innovations in solid-phase oligonucleotide chemistry, particularly work carried out by Robert L. Letsinger at Northwestern University and by Marvin H. Caruthers and colleagues at the University of Colorado Boulder (Egli, 2023; Beaucage and Caruthers, 1981).

Robert L. Letsinger, a chemist at Northwestern University from 1946 until his retirement in 1991, made foundational contributions to synthetic DNA chemistry (Egli, 2023). His work helped establish several central steps in oligonucleotide synthesis, including the solid-phase approach, the phosphotriester method and the P(III) phosphite–triester method. These developments were later refined into the phosphoramidite technology that became central to automated DNA and RNA synthesis (Egli, 2023).

A major technical challenge in early oligonucleotide synthesis was the efficient formation of internucleotide phosphodiester linkages. Letsinger

and co-workers introduced the phosphite coupling approach, in which protected nucleosides were reacted through phosphorus(III) intermediates and subsequently oxidized to form natural phosphate internucleotide bonds (Beaucage and Caruthers, 1981). This strategy represented an important advance because it allowed more efficient coupling than earlier approaches; however, the reactive intermediates, such as nucleoside phosphomonochloridites and monotetrazolides, were unstable toward hydrolysis and air oxidation (Beaucage and Caruthers, 1981). As a result, the reactive species often had to be prepared immediately before use or stored under restrictive conditions, limiting the practicality of the method for routine or automated synthesis (Beaucage and Caruthers, 1981).



In 1981, Serge L. Beaucage and Marvin H. Caruthers, working in the Department of Chemistry at the University of Colorado, Boulder, reported a new class of deoxynucleoside phosphoramidite intermediates (Beaucage and Caruthers, 1981). Their paper described N,N-dimethylaminophosphoramidites of protected deoxynucleosides as stable compounds that could be prepared by standard organic chemistry procedures and stored as dry powders under normal laboratory conditions (Beaucage and Caruthers, 1981). This solved a central limitation of earlier phosphite chemistry: the instability of the activated phosphorus intermediates. Caruthers demonstrated that these phosphoramidites could be activated under mild acidic conditions, particularly using 1H-tetrazole, and could condense with protected nucleosides to form internucleotide phosphite linkages in essentially quantitative yield before oxidation to the natural phosphate form (Beaucage and Caruthers, 1981).

Beaucage and Caruthers also showed that the new reagents were compatible with polymer-supported synthesis, an essential requirement for automated oligonucleotide production. In trial reactions using nucleoside-functionalized silica gel, the phosphoramidite reagents produced dinucleotides including d(TpG), d(CpG), d(ApG) and d(GpG) in high yields of 100%, 98%, 94% and 93%, respectively, after oxidation and deprotection

(Beaucage and Caruthers, 1981). These results demonstrated that phosphoramidite chemistry could support efficient stepwise chain extension on a solid support, providing the chemical foundation for automated oligonucleotide synthesis and, later, for the scalable production of ASOs (Beaucage and Caruthers, 1981; Egli, 2023).

By the late twentieth century, oligonucleotide chemistry had moved from a specialized synthetic process to an industry standard. Letsinger's early work at Northwestern established key principles of solid-phase and phosphite-based synthesis, while Beaucage and Caruthers' work at Colorado provided stable phosphoramidite building blocks that made high-yield, routine and automated synthesis feasible (Beaucage and Caruthers, 1981; Egli, 2023). These developments made it possible to prepare cost-effective ASOs with defined sequence, length and chemical composition, enabling their later application as research tools and therapeutic agents.

Automation of ASO synthesis

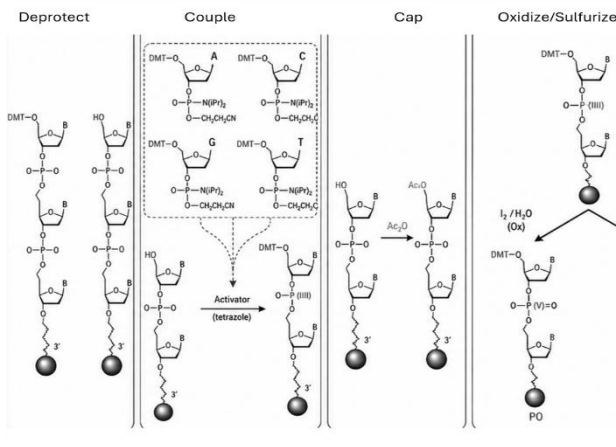
The automation of antisense oligonucleotide synthesis developed from the earlier transition between manual phosphoramidite chemistry and computer-controlled solid-phase oligonucleotide synthesis. Once Beaucage and Caruthers introduced stable deoxynucleoside phosphoramidite intermediates in 1981, oligonucleotide synthesis became suitable for repeated, programmed chemical cycles because each nucleotide addition could be carried out through a predictable sequence of reactions on a solid support (Beaucage and Caruthers, 1981). This chemical reliability was essential for automation; a synthesizer depends on the reproducibility of each cycle of 1) deprotection, 2) coupling, 3) capping and 4) oxidation. This four-step cycle is the basis of modern column-based oligonucleotide synthesis, in which nucleotide monomers are progressively added to solid-phase supports such as controlled pore glass or polystyrene beads.

In automated column-based synthesis, the growing oligonucleotide chain is attached to a solid support inside a synthesis column. A computer-controlled system then delivers reagents in a fixed order. First, the protecting group on the terminal nucleotide is removed during deprotection; second, the next phosphoramidite monomer is coupled to the growing chain; third, unreacted chains are capped to prevent them from participating in later cycles; and fourth, the newly formed phosphite linkage is oxidized to the more stable phosphate linkage (Ma et al., 2024). After the desired sequence has been assembled, the oligonucleotide is cleaved from the support under alkaline conditions and purified by methods such as oligonucleotide purification columns, polyacrylamide gel electrophoresis or high-performance liquid chromatography (Ma et al., 2024).

The emergence of automated DNA synthesizers in the 1980s therefore represented a practical turning point in ASO production. Ma et al. (2024) states that Applied Biosystems developed one of the first automated DNA synthesizers using the column-based method, improving the accessibility of synthetic oligonucleotides. These instruments used synthesis columns as support and computer-programmed solenoid valves to control the flow of reagents into the column, enabling the programmed synthesis of single-stranded DNA (Ma et al., 2024). For ASO synthesis, this automation was especially important because antisense molecules must be produced with a defined sequence, length and chemical composition. Manual synthesis made this possible only on a limited scale, whereas automated synthesis made routine and reproducible production feasible.

Automation also improved the scale and throughput of oligonucleotide production. Commercial column-based instruments, including the MerMade series from BioAutomation and the Dr. Oligo series from Biolytic, are identified by Ma et al. (2024) as widely used examples of automated oligonucleotide synthesis platforms. The same review notes that modern automated column-based synthesis can support throughput ranging from 96 to 1536 oligonucleotides, while reducing the cost of sequences under 100 bases to approximately US\$0.05–0.15 per nucleotide and lowering error rates to approximately one mutation per 200 base pairs or less (Ma et al., 2024). These improvements were significant for ASOs because therapeutic and research-grade antisense molecules require consistent quality, controlled impurities and scalable production.

The automation of ASO synthesis can therefore be understood as both a chemical and engineering achievement. Chemically, phosphoramidite synthesis supplied stable monomers and efficient coupling reactions. Technologically, automated synthesizers supplied timed reagent delivery, controlled washing steps and repeatable reaction cycles. Together, these advances allowed synthetic oligonucleotides to move from specialist laboratory products to widely available research, diagnostic and pharmaceutical materials. Ma et al. (2024) explicitly links the automation and commercialization of phosphoramidite chemistry in the 1980s with the breakthrough development of oligonucleotide synthesis, noting that oligonucleotides are now used across scientific research, clinical research, pharmaceutical development and molecular diagnostic probe synthesis.



Automated phosphoramidite synthesis also has technical limitations. Although it is mature and reliable, the synthesis of oligonucleotides longer than approximately 200 bases remain difficult; even very high per-cycle efficiencies produce accumulated errors across many cycles. The practical synthesis limit is often closer to 100 bases, depending on cost and quality requirements (Ma et al., 2024). ASO manufacturing still faces challenges associated with reagent cost, organic solvent use, toxic chemical waste and the need for purification after synthesis (Ma et al., 2024).

More recent forms of automation have expanded beyond conventional single-column synthesis. Microarray-based oligonucleotide synthesis allows large numbers of distinct oligonucleotides to be produced in parallel on silicon chips or glass slides. According to Ma et al. (2024), these platforms use methods such as photolithography, electrochemical synthesis and ink-jet printing to generate large numbers of oligonucleotide sequences while reducing reagent use and cost. For example, the review describes photolithographic platforms capable of synthesizing up to 500,000 oligonucleotides on a single chip, ink-jet systems that can produce hundreds of thousands to over one million oligonucleotides in parallel, and electrochemical systems capable of synthesizing millions of unique oligonucleotides (Ma et al., 2024). Although these methods are more commonly associated with oligonucleotide libraries and gene synthesis rather than individual therapeutic ASO manufacture, they show how automation has shifted the field toward high-throughput, miniaturizing and digitally controlled synthesis.

Enzymatic Synthesis

Instead of relying on chemical coupling, enzymatic approaches use polymerases, ligases or enzyme cascades to form phosphodiester bonds under milder, usually aqueous, conditions (Pichon and Hollenstein, 2024; Zhou et al., 2025). Polymerase-mediated synthesis can, in principle, generate much longer sequences than chemical synthesis, as shown by in vitro transcription methods used to produce long modified mRNA molecules (Pichon and Hollenstein, 2024). However, natural polymerases are usually template-dependent, meaning they copy an existing nucleic acid template rather than freely writing an arbitrary user-defined sequence. This creates a central challenge: enzymatic synthesis must be made programmable if it is to compete with the sequence control of phosphoramidite chemistry.

While controlled enzymatic DNA synthesis is relatively advanced, RNA synthesis remains more difficult. The key structural complication is the presence of both 2'-OH and 3'-OH groups in ribonucleotides. Blocking only the 3'-OH can allow later incorporation of therapeutically relevant 2' modifications, but selective 3' protection is synthetically challenging. Blocking the 2',3'-cis-diol is easier and can reduce unwanted 2'-5' branching, but it limits access to 2'-modified therapeutic RNA structures (Pichon and Hollenstein, 2024).

Recent work has shown that controlled enzymatic RNA synthesis is possible using suitable combinations of reversible terminator nucleotides and template-independent polymerases such as poly(U) polymerase (PUP) and poly(A) polymerase (PAP). Pichon and Hollenstein (2024) describe the use of 3'-O-allyl and 3'-O-propargyl ribonucleotides with engineered PUP variants, achieving high coupling yields across multiple synthetic cycles. Similarly, Zhou et al. (2025) reports that CID1 poly(U) polymerase mutants can support aqueous, stepwise RNA synthesis using reversible terminator nucleotides bearing 3'-O-allyl ether blocking groups. Together, these studies suggest that enzymatic RNA synthesis is moving beyond proof-of-principle toward a credible platform technology. Despite this progress, the approach remains less mature than enzymatic DNA synthesis and far less established than conventional phosphoramidite chemistry.

XNA and chemically modified oligonucleotides

A major future goal is the enzymatic synthesis of chemically modified oligonucleotides, including xeno nucleic acids (XNAs), locked nucleic acids (LNAs), threose nucleic acids (TNAs), hexitol nucleic acids (HNAs), phosphorothioates and 2'-modified RNAs. These modifications are especially important in therapeutic oligonucleotides because they can